

1 What is a VIEW?

Definition

A **VIEW** is a:

- 👉 **Virtual table**
 - 👉 Created using a **SELECT** query
 - 👉 Does NOT store actual data (in most cases)
-

Simple Understanding

- 👉 “View = Saved Query”

Instead of writing complex JOIN again and again:

- 👉 Save it as a VIEW
 - 👉 Use like a table
-

2 Example (Very Important)

Tables:

- customers
 - orders
-

Create View

```
CREATE VIEW customer_orders AS  
SELECT c.cust_name, o.order_id, o.order_date  
FROM customers c  
JOIN orders o ON c.cust_id = o.cust_id;
```

Use View

```
SELECT * FROM customer_orders;
```

- 👉 Works like a normal table ✅

3 Why Use Views?

Security

👉 Hide sensitive columns

```
CREATE VIEW safe_customers AS  
SELECT cust_name, city  
FROM customers;
```

Simplicity

👉 Hide complex JOIN logic

Reusability

👉 Write once, use many times

Data Abstraction

👉 Show only required data

4 Types of Views

1. Simple View

👉 Based on single table

```
CREATE VIEW simple_view AS  
SELECT cust_name, city FROM customers;
```

2. Complex View

👉 Uses JOIN, GROUP BY

```
CREATE VIEW order_summary AS
SELECT c.cust_name, COUNT(o.order_id)
FROM customers c
JOIN orders o ON c.cust_id=o.cust_id
GROUP BY c.cust_name;
```

5 Create View Syntax

```
CREATE VIEW view_name AS
SELECT columns FROM table;
```

6 Replace View

```
CREATE OR REPLACE VIEW view_name AS
SELECT columns FROM table;
```

7 Drop View

```
DROP VIEW view_name;
```

8 Update Through View (Important 🔥)

👉 Possible only if:

- Based on single table
 - No GROUP BY
 - No DISTINCT
-

Example

```
UPDATE simple_view
SET city='Pune'
WHERE cust_name='Amit';
```

9 When View is NOT Updatable

✗ Contains:

- GROUP BY
 - JOIN (most cases)
 - DISTINCT
 - Aggregate functions
-

10 WITH CHECK OPTION

👉 Ensures inserted data satisfies view condition

Example

```
CREATE VIEW pune_customers AS
SELECT * FROM customers
WHERE city='Pune'
WITH CHECK OPTION;
```

👉 Now you cannot insert non-Pune data ✗

11 Materialized View (Concept)

👉 Stores actual data (not in MySQL by default)

View Type	Stores Data
Normal View	✗ No

Materialized View ☒ Yes

12 View vs Table

Feature	View	Table
Stores data	✗	✓
Virtual	✓	✗
Fast query	✗ (depends)	✓
Security	✓	✗

13 Real-Life Example (Company)

👉 HR should not see salary:

```
CREATE VIEW employee_public AS  
SELECT emp_name, department  
FROM company_data;
```

14 Common Mistakes

- ✗ Thinking view stores data
- ✗ Updating complex view
- ✗ Forgetting CHECK OPTION
- ✗ Using view instead of index for performance